

Participant & Site Burden Resource Landscape for Protocol Optimization

Optimizing Data Collection Initiative

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Disclaimer

This document contains a list of resources currently available related to measuring participant and site burden related to data collection in clinical trials. The list is provided as is and for informational purposes only.

The list was pulled together by TransCelerate's Optimizing Data Collection initiative, which has collaborated with Tufts CSDD on new benchmarking measures for data collection. The initiative seeks to encourage sponsors to consider participant and site burden related to data collection and design trials accordingly.

With the exception of the TransCelerate tools on slide 13, TransCelerate has not had any involvement with the development and/or use of these tools. TransCelerate is not endorsing and does not endorse any tool, vendor or industry group. TransCelerate has not evaluated the tools developed by others and makes no representations or warranties as to whether they are fit for purpose. The descriptions listed are obtained directly from the solution developer's website. Companies choose which resources they will use, if any, whether on this list or not, in their sole discretion.

Introduction: ODC Initiative Background

Increasing complexity in clinical trial design has contributed to unnecessary burden on participants, sites, and the broader ecosystem. Evidence from TransCelerate's collaboration with the Tufts Center for the Study of Drug Development¹ highlights that protocol complexity and data collection practices are key drivers of this burden, reinforcing the need for more intentional and targeted approaches to protocol design.

This need for optimization is further reinforced by evolving ICH guidance, which emphasizes quality-by-design and fit-for-purpose data collection as core expectations for modern clinical trials.

In response, the Optimizing Data Collection (ODC) initiative supports sponsors in thinking more critically about what data is collected—and why—by providing structured approaches to evaluate which procedures and data points are needed and how often they should be captured.

There is a range of tools, frameworks, guidance that aim to address aspects of protocol design burden. However, these resources may not be widely established across the broader ecosystem, resulting in inconsistent application and limited integration.

To address this gap, the TransCelerate Optimizing Data Collection (ODC) initiative developed this resource pack to enhance visibility to existing participant and site-burden related resources, consolidating them into a single, easy-to-navigate reference.

¹[*Insights Informing Strategies for Optimizing the Collection of Clinical Trial Data, Tufts CSDD, 2025*](#)

Introduction: Resource Pack Structure and Navigation

How the Resource is Structured

Each resource is presented in a consistent tabular format to enable quick scanning and side-by-side visibility across identified resources:

- Authoring Organization
- Resource Name
- Brief Description – Direct from Source
- Access Information

This structure is intended to support efficient navigation and a high-level view of available resources in the context of protocol design activities.

Understanding Resource Categories

Resources are organized into three primary categories:

1. Vendor Resources & Tools
2. Industry Group Resources & Tools
3. Peer-Reviewed Literature

Within each category, materials are further grouped based on the type of organization that developed the resource.

These categories are intended to help better understand the origin and perspective of each resource. They are not mutually exclusive and do not imply prioritization, endorsement, or guidance on how resources should be selected or applied.

Introduction: Intended Impact and Scope

Intended Impact

The intent of this resource pack is two-fold:

- **Increase awareness** of existing tools and approaches across the clinical research ecosystem
- **Encourage sponsors to consider participant and site burden** related to data collection and design trials accordingly

Scope and Limitations

This compilation reflects a landscape assessment as of June 2026. While not exhaustive, it offers a curated view of tools and resources to help understand the impact of protocol complexity on participant and site burden – particularly at the intersection of data collection and trial design – and to support assessment of protocol data collection. While approaches and methodologies vary, this resource prioritizes solutions aligned to this focus area.

We recognize that many additional resources exist across the industry. As part of this effort, we actively sought input through broad engagement and crowdsourcing, incorporating perspectives from a range of solution providers and stakeholders. We will continue to incorporate additional input and refine this resource, as appropriate, through year-end 2026 as relevant additions are identified.

If you believe a resource should be considered, please email the Optimizing Data Collection initiative at ODC@transceleratebiopharmainc.com for review and potential inclusion.

Reference Table of Vendor Resources & Tools



Vendor Resources & Tools (1/3)

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
Evidera Thomas C et al.	How do study design features and participant characteristics influence willingness to participate in clinical trials? Results from a choice experiment (2022)	This study quantifies how study designs influence willingness to participate and how this varies with participant types. These findings suggest that it is how an indication influences quality of life and treatment experience, rather than the indication alone, that impacts participation rates, opening the way for insights that are transferable across indications, which may be particularly useful when considering rare diseases.	Access Here BMC Medical Research Methodology
Evinova	Evinova AI	Evinova's AI platform accelerates clinical development while giving sponsors control, compliance, and confidence across every phase of the trial lifecycle.	Access Here
Faro Health	Faro Health Study Designer	The Faro Health Study Designer empowers teams to construct and compare multiple study designs in order to strike a balance between scientific complexity and patient centrality. Cycle times see improvement with real-time feasibility insights into patient burden, site burden, cost implications, as well as study complexity.	Access Here
Faro Health	Reducing Patient Burden in Clinical Trials by Quantifying It (Blog)	At Faro, we believe one of the biggest hurdles to reducing patient burden is being able to quantify it. Using Faro's Study Designer, clinical development teams can get real-time insights into the impact of their specific protocol on patient burden during the protocol planning process itself.	Access Here
Faro Health and Merck & Co. Cummings SR et al.	A Method to Redesign and Simplify Schedules of Assessment and Quantify the Impacts. Applications to Merck Protocols (2024)	We developed an approach, termed Lean Design, for redesigning the assessments in SoAs that generate data, the 'Data SoA.' It starts with a simple "ground zero" SoA. Any addition is challenged using several principles of trial design. We employed a system, the Faro Trial Designer Tool, to quantify the impacts of changes in an SoA to provide real-time feedback to the team and sponsor. We applied the approach in workshops with teams for six clinical trials in various stages of design and implementation.	Access Here Therapeutic Innovation & Regulatory Science

Vendor Resources & Tools (2/3)

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
IQVIA	IQVIA Data-Informed Protocol Assessment	IQVIA's Data-Informed Protocol Assessment (DIPA) helps you improve protocols by evaluating key areas of potential impact, such as patient burden and design inconsistencies. DIPA analytics validate your design decisions and identify potential study risks prior to protocol approval.	Access Here
IQVIA	Empowering Clinical Research Sites and Sponsors in the Patient-centric Era (White Paper)	Strategic site collaboration can foster patient-centricity by recognizing challenges faced by clinical trial sites and redefining the site as the hub of patient-centric trials. With collaboration instead of delegation, sponsors can decrease site challenges and maintain patient engagement as the industry continues to shift toward a more patient-centric model.	Access Here
Medidata	Medidata Platform	The Medidata Platform aligns the Patient, Data, and Study Experiences into one continuous, governed system, so you accelerate start-up, strengthen data quality, and deliver faster, more approval-ready submissions at scale.	Access Here
Medidata and Novo Nordisk	An Innovative Approach to Patient-centric and Data-driven Clinical Trial Planning using the Patient Burden Index (White Paper)	In the interest of increasing patient centricity in clinical trials, Acorn AI and Novo Nordisk have conducted extensive research on the correlation between the Patient Burden Index and clinical trial operational performance. Our joint analysis confirms the intuition that a high PBI is associated with lower enrollment rates and lengthier timelines to reach the milestone of 25% target enrollment.	Access Here
Phesi	Trial Accelerator	A self-service SaaS platform giving life sciences companies access to the power of the world's largest clinical trials database containing real-world data. Delivering indication specific predictive modelling across patient population identification, trial performance, country, investigator and site selection.	Access Here

Vendor Resources & Tools (3/3)

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
PPD (Thermo Fisher Scientific Inc.)	StudyGage	StudyGage, an innovative, proprietary study patient burden tool, leverages more than 60,000 patient study participation decisions collected from a diverse sample of patients from around the world. In a few simple steps, StudyGage provides access to validated assessments of how standard and alternative study designs will impact patient participation decisions.	Access Here
PPD (Thermo Fisher Scientific Inc.)	Addressing the challenges of conducting qualitative interviews within clinical trials (White Paper)	Incorporating qualitative interviews within clinical trials can add meaningful value, but they also pose unique operational and logistical challenges that require careful planning and preparation. This white paper provides insights and recommendations for trial sponsors to successfully address four key areas where significant challenges occur when implementing qualitative research within the context of a clinical trial, and details our model for conducting interviews within clinical trials.	Access Here
Sage Arbor	Schedule of Assessments Optimizer (Open Source)	An intelligent tool designed to streamline clinical trial protocols by analyzing and optimizing the Schedule of Assessments for patient and site burden.	Access Here
WCG	ClinSphere Trial IntelX™	Every protocol choice – from visit schedules to eligibility criteria – impacts how a trial performs. WCG ClinSphere Trial IntelX™ gives sponsors, CROs, and sites the power to predict those outcomes before first participant in. Built on 80,000+ complete protocols and 40,000+ operational, performance-benchmarked trials, Trial IntelX uses AI and real-world performance and operational data to forecast timelines, enrollment, and site and participant burden with precision.	Access Here
ZS	Trials.ai Smart Designer	Smart Designer empowers teams to build optimized clinical studies and generate protocol documents with AI-driven recommendations, benchmark comparisons, and real-time tradeoff analysis. From study setup to protocol draft, streamline the entire workflow.	Access Here

Reference Table of Industry Group Resources & Tools



Industry Group Resources & Tools (1/2)

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
Center for Information and Study on Clinical Research Participation (CISCRP)	CISCRP's Perceptions and Insights Study	The Perceptions and Insights Study has been conducted since 2013 and helps establish a routine global assessment of trends in clinical research public perceptions. Industry stakeholders can use these reports to identify opportunities to better inform and engage all stakeholders and partners in the clinical research enterprise. CISCRP receives input and support from biopharmaceutical companies, contract research organizations, industry service providers, patient advocates/advocacy groups, and investigative sites to execute the study.	Access Here
International Children's Advisory Network (iCAN)	International Position Paper on Youth Partnership in Healthcare	This position paper was co-developed by 76 young people aged 12 to 23 from 11 countries, during the 2025 iCAN (International Children's Advisory Network) Summit. Through a co-construction process with peers and adult allies, youth identified four key priorities to strengthen the development of meaningful partnerships in health with young people. Their voices reflect diverse realities, but a shared goal: to be recognized as active contributors to the health systems. The paper is a call to action for leaders, professionals, and institutions to build sustainable partnerships with youth on health-related matters.	Access Here
Society for Clinical Research Sites (SCRS)	2025 Site Landscape Survey Report	The SCRS Site Landscape Survey serves as a critical barometer of industry conditions by gathering comprehensive data about site operations, financial sustainability, technology adoption, workforce challenges, and relationships with sponsors and CROs.	Access Here

Industry Group Resources & Tools (2/2)

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
TransCelerate Biopharma, Inc. (TCB)	Patient Protocol Engagement Toolkit (P-PET)	A comprehensive set of materials that sponsors and other stakeholders can use to engage patients during protocol development. The goal is to improve patient experience and reduce patient burden as a study participant. The toolkit now includes caregiver considerations, engaging in a virtual environment, patient technology, and diversity and inclusion.	Access Here
	Personalized Clinical Trial Framework	A set of practical considerations to support the R&D ecosystem (i.e., sponsors, sites, third parties, and other clinical trial stakeholders) with understanding and evaluating options that enable a more personalized experience for clinical trial participants across diverse populations.	Access Here
	Study Feedback Participant Questionnaire (SPFQ) Toolkit	A survey provided to study participants at the beginning, middle, and end of a clinical study. Translations of the SPFQ are available in 15+ languages. A Question Bank Addendum was added to include additional questions focused on motivation to participate, gratitude, and decentralized clinical trial activities. The toolkit also includes a Socialization Presentation, Implementation User Guide, and FAQ.	Access Here

Reference Table of Peer-Reviewed Literature

Peer-Reviewed Literature

Approaches to Measuring and Assessing Burden (1/2)

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
Getz K et al. TransCelerate Biopharma Inc. and Tufts CSDD	Insights Informing Strategies for Optimizing the Collection of Clinical Trial Data (2025)	Tufts CSDD and 15 TransCelerate Biopharma sponsor companies convened to update benchmarks and gather new insights into opportunities to optimize clinical trial data collection. In all, 105 multi-therapeutic protocols with a primary completion date after 2018 were analyzed. Data volume by study and by participant were analyzed and procedures were categorized into core, standard and non-core based on the endpoints they supported.	Access Here Therapeutic Innovation & Regulatory Science
Dirks A et al. Tufts CSDD	Trends in the Burden for Patients Participating in Industry-Funded Clinical Trials (2025)	Through individual and larger group studies, the Tufts Center for the Study of Drug Development (Tufts CSDD) has collaborated with 14 pharmaceutical companies in applying a validated approach early in the protocol design process to assess patient participation burden. Data from these assessments has been aggregated to analyze trends in participation burden between 2011 and 2022.	Access Here Therapeutic Innovation & Regulatory Science
Dirks A et al. BMS, ZS and Tufts CSDD	Measuring Patient Participation Burden in Clinical Outcome Assessments for Clinical Trials (2025)	Bristol Myers Squibb (BMS), in collaboration with ZS and Tufts CSDD, recently conducted a detailed evaluation of participation burden associated with clinical outcome assessments, including patient reported outcomes.	Access Here Therapeutic Innovation & Regulatory Science
Aiyegbusi OL et al. International multi-stakeholder group	Recommendations to address respondent burden associated with patient-reported outcome assessment (2024)	We conducted an initial literature review to generate a list of candidate recommendations aimed at reducing respondent burden. This was followed by a two-stage Delphi survey by an international multi-stakeholder group. A consensus meeting was held to finalize the recommendations. The final consensus statement includes 19 recommendations to address PRO respondent burden in healthcare research and clinical practice. If implemented, these recommendations may reduce PRO respondent burden.	Access Here Nature Medicine
Klüglic M et al. Boehringer Ingelheim and FindMeCure	Patient feasibility as a novel approach for integrating IRT and LCA statistical models into patient-centric qualitative data—a pilot study (2024)	This pilot study aims to explore and demonstrate the analytical framework of the "patient feasibility" concept—a novel approach for integrating patient-centric data into clinical trial design using psychometric latent class analysis (LCA) and interval response theory (IRT) models.	Access Here Frontiers

Peer-Reviewed Literature

Approaches to Measuring and Assessing Burden (2/2)

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
Florez M et al. Tufts CSDD	Quantifying Site Burden to Optimize Protocol Performance (2024)	The Tufts Center for the Study of Drug Development conducted an online survey among clinical research sites worldwide and gathered 355 responses assessing the burden associated with distinct activities and procedures related to the implementation of clinical trial protocols using traditional and decentralized approaches.	Access Here Therapeutic Innovation & Regulatory Science
Smith Z et al. Tufts CSDD and ZS	Insights from a Multi-company Workshop to Apply a Patient Participation Burden Algorithm to Protocol Data (2022)	Utilizing a participation burden algorithm developed in a previous study, Tufts CSDD, in collaboration with ZS, led a workshop among 8 pharmaceutical companies to validate the methodology of benchmarking the participation burden of a set of retrospective protocols and comparing these data to a prospective protocol design.	Access Here Therapeutic Innovation & Regulatory Science
Smith Z et al. Tufts CSDD and ZS	Enhancing the Measure of Participation Burden in Protocol Design to Incorporate Logistics, Lifestyle, and Demographic Characteristics (2021)	This study builds on prior work conducted by the Tufts Center for the Study of Drug Development in collaboration with ZS. A comprehensive participant burden algorithm based on protocol procedures, participation requirements and lifestyle preferences was developed and tested. Clinical trial preferences and perceptions from 3002 global patients were analyzed to inform and derive the algorithm. It was next tested against a convenience sample of 266 completed protocols. Descriptive statistics, significance tests, and regression analyses were performed.	Access Here Therapeutic Innovation & Regulatory Science
Getz K et al. Tufts CSDD	Assessing Patient Participation Burden Based on Protocol Design Characteristics (2020)	This paper presents the results of a cross-sectional pilot study conducted by the Tufts Center for the Study of Drug Development and ZS Associates among nearly 600 patients via an online survey conducted between February and March 2019. Respondents rated the perceived burden of 60 commonly administered protocol procedures.	Access Here Therapeutic Innovation & Regulatory Science

Peer-Reviewed Literature

Drivers of Burden in Protocol Design

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
Abdel-Rahman S et al. Pediatric Trials Network Steering Committee	Sponsor- vs. FDA-Initiated Changes to Pediatric Clinical Trial Protocols: A Comparison of Associated Participant Burden (2025)	Half of all protocol amendments are associated with changes that conceivably alter the burden of clinical trial participation, and these changes are twice as likely to add to (vs. reduce) participant burden. Given the time and effort involved with protocol amendments, we suggest that modifications to trial protocols be accompanied by a reassessment of the role and utility of all existing protocol elements at the time of resubmission.	Access Here Therapeutic Innovation & Regulatory Science
Smith Z et al. Tufts CSDD	A Case Study Assessment on the Rationale for, and Relevance of, Non-Core Protocol Data (2023)	To better understand the nature of Non-Core procedures and derive new insight into protocol simplification and optimization, Tufts CSDD collaborated with the FDA and sponsor companies to assess alignment on the rationale for collecting, and relevance of, Non-Core protocol data. Twelve sponsor companies classified and rated 700 distinct procedures from 19 pivotal trials supporting new drug and biologics approvals. FDA reviewers classified and rated 80 distinct procedures for three of the 19 pivotal trials. The results of this assessment indicate areas of alignment and misalignment.	Access Here Therapeutic Innovation & Regulatory Science
Getz K et al. Tufts CSDD	Quantifying the magnitude and cost of collecting extraneous protocol data (2015)	Between November 2011 and May 2012, Tufts Center for the Study of Drug Development conducted a study among a working group of 15 pharmaceutical companies in which a total of 25,103 individual protocol procedures were evaluated and classified using clinical study reports and analysis plans.	Access Here American Journal of Therapeutics

Peer-Reviewed Literature

Burden Across Populations and Settings

Authoring Organization	Resource Name	Brief Description – Direct from Source	Access
Baba A et al. Standards for Research (StaR) in Child Health	Developments in the Design, Conduct, and Reporting of Child Health Trials (2024)	As a supplement to recent scoping reviews and analysis of the literature, the organizers (A.B., T.K., M.O.) aimed to conduct a review of progress and new directions in child health trials' methods developments since the publication of the StaR Child Health standards in 2012 in Pediatrics.	Access Here Pediatrics
Domeisen Benedetti F et al.	Self-Assessment Instruments for Supporting Family Caregivers: An Integrative Review (2024)	The aim of the present study was to identify and appraise self-assessment instruments (SAs) that capture the dimensions of family caregivers' burdens and that support family caregivers in easily identifying their caregiving role, activities, burden, and needs.	Access Here Healthcare
Silva N et al.	The Applicability of the "Revised Burden Measure" in Pediatric Settings: Measuring Parents' Caregiving Burdens and Uplifts (2020)	The aim of this study was to systematically examine the psychometric properties of the "Revised Burden Measure" in a sample of parents of children with and without chronic health conditions. The "Revised Burden Measure" and self-reported measures of quality of life and mental health were administered to a sample of parents who had children with (n = 277) or without (n = 117) chronic health conditions.	Access Here Journal of Child and Family Studies
Naidoo N et al.	The research burden of randomized controlled trial participation: a systematic thematic synthesis of qualitative evidence (2020)	Participation in randomized controlled trials (RCTs) may be quite demanding and could represent an important burden for patients. We aimed to explore this research burden (i.e., the psychological, physical, and financial burdens) experienced by patients through their participation in a RCT. Findings from 45 qualitative studies that involved 1732 RCT participants provide a detailed description of research burden across the whole RCT process.	Access Here BMC Medicine